

L13 - Linearity and Superposition

17 March 2024 15:52

Core Principles: Ideal circuits

Linearity

- Ratio between any two voltages or currents in a circuit is constant

$$\frac{V_1}{V_2} = c \quad \frac{V_1}{V_2} = \frac{5}{2} = 2,5$$

$$\frac{I_1}{I_2} = c \quad \frac{I_1}{I_2} = \frac{1,5}{0,6} = 2,5$$

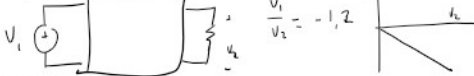
Superposition

- Each independent source contributes to the voltage/current across an element
- The voltage/current is the sum of the contributions of each independent source
- $V_{out} = V_{cont1} + V_{cont2} + V_{cont3} \dots$
- Turn off voltage source - replace with short circuit
- Turn off current source - replace with open circuit
- Never turn off dependant sources

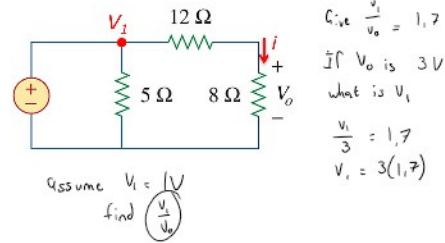
$$V_1 \rightarrow 20\% \rightarrow V_2 \rightarrow 2V$$

$$V_1 \rightarrow 80\% \rightarrow V_2 \rightarrow 8V$$

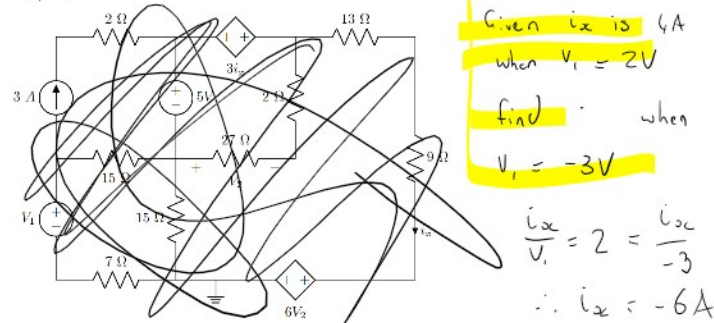
Why use linearity?



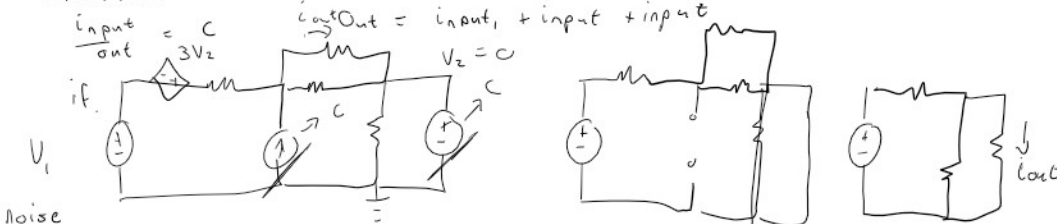
Example 1:



Example 2:



Why use superposition?



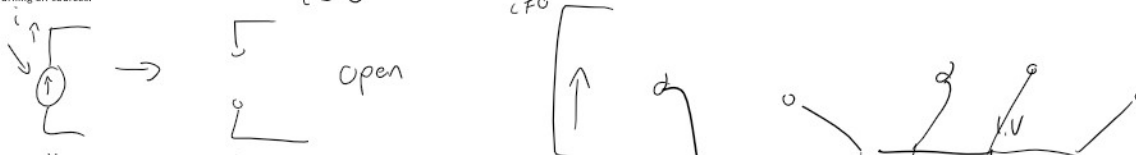
noise

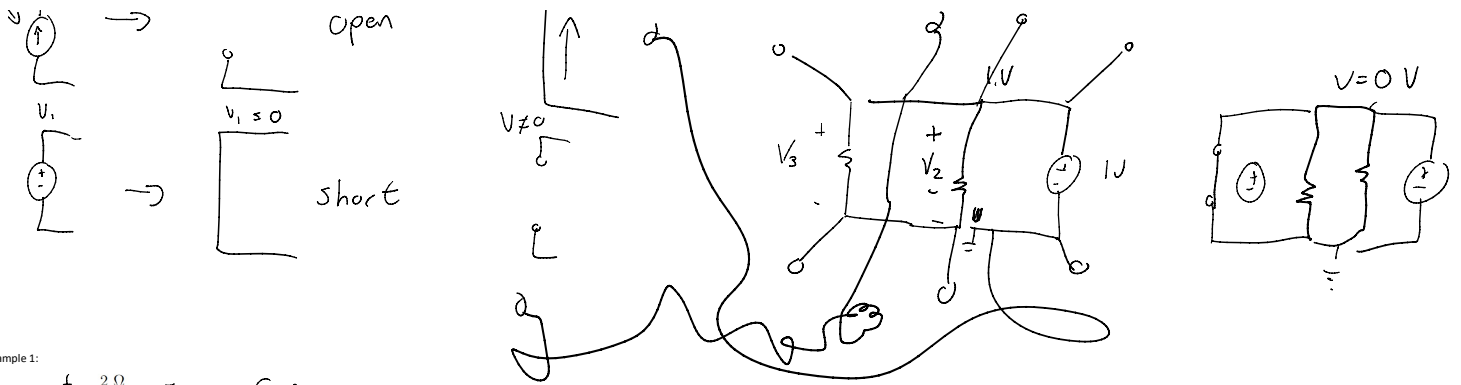
Steps:

- Turn off all independent sources except test source
- Solve for desired answer
- Repeat for all independent sources
- Add to get final answer

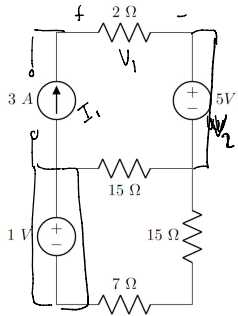
never turn it off

Turning off sources:



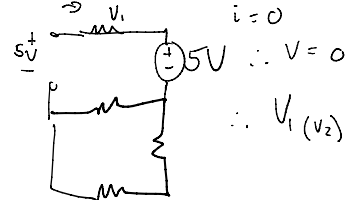


Example 1:



① find independent contribution

of 5V source to V_1 :



$$V_1(1V) = 2V$$

$$\therefore V_1 = V_1(1V) + V_1(I_x) + V_1(V_2)$$

$$\therefore V_1 = 3.2$$

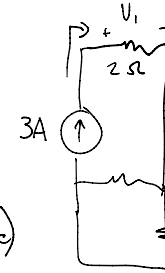
$$= 6V = 2 + 6 + 0$$

$$= 8V$$

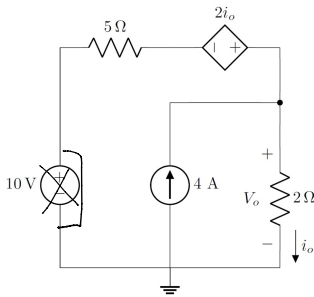
$$V_1(I_x) = 6V$$

② find contribution of I_1 (current source) to V_1 ?

$$V_1(I_x)$$

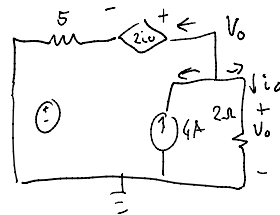


Example 2:



find the contribution of 4A source

to i_0 :



$$\frac{V_0 - 2i_0}{5} + (-4) + \frac{V_0}{2} = 0$$

$$i_0 = \frac{V_0}{2}$$

$$\frac{V_0 - 2\left(\frac{V_0}{2}\right)}{5} - 4 + \frac{V_0}{2} = 0$$

$$\frac{V_0}{2} = 4$$

$$i_0 = 4A$$

$$V_0 = 8$$

$$i_0(4A) = 4A$$